

PROJECT ABSTRACT

Why

The purpose of this project is to understand the factors that influence the popularity of YouTube videos. It focuses on identifying how engagement metrics such as views, likes, and comments contribute to a video's success. The study also examines the role of video categories in determining trending behavior. Additionally, publish time is analyzed to understand when videos are more likely to gain audience attention. This project helps in exploring user interaction patterns on YouTube. Overall, it aims to explain what makes a video trend on the platform.

What

This project analyzes a YouTube Trending Videos dataset using data analysis and machine learning techniques. The dataset includes important features such as video titles, views, likes, dislikes, comments, and tags. It also contains information about video categories and publish time. These attributes are used to study patterns in trending videos. Machine learning models are applied to find relationships between different variables. The main goal is to predict video popularity and trending status.

How

The project begins with data cleaning, where missing values and duplicates are handled. After that, feature engineering is performed by creating new variables like publish hour and engagement rate. Exploratory Data Analysis (EDA) is conducted using visualizations to understand data patterns. Relationships between views, likes, comments, and categories are analyzed. Machine learning models such as Linear Regression and Random Forest Regression are used for prediction. Finally, Logistic Regression is applied to classify whether a video will trend or not.

PROBLEM STATEMENT

Understanding what factors make YouTube videos trend and predicting video popularity using machine learning.

METHODOLOGY

Step-by-Step workflow:

1. Data collection
2. Data cleaning
3. Exploratory Data Analysis (EDA)
4. Visualization
5. Feature selection
6. Machine learning model training
7. Prediction and evaluation

INTRODUCTION TO THE PROJECT

This project analyzes YouTube trending videos to understand what factors affect video popularity using views, likes, comments, category, and publish time.

The goal of this project is to analyze YouTube trending videos and identify patterns that make videos popular. Machine learning models are used to predict view counts and trending status.

DATASET DESCRIPTION

The dataset used in this project is based on YouTube Trending Videos. It contains information about videos that appeared in YouTube's trending section. The main purpose of this dataset is to analyze video popularity and understand which factors help videos become trending. The dataset includes different types of information related to videos, channels, audience engagement, and publishing details. Each row in the dataset represents one YouTube video, while each column represents a specific feature or attribute of that video.

The dataset contains important features such as video title, video category, channel name, views, likes, dislikes, comments, tags, and publish time. These features help in understanding audience behavior and measuring video performance on YouTube. Views represent the number of times a video has been watched. Likes and

comments show audience interaction and engagement with the video. The publish time feature helps analyze how upload timing affects video popularity. Categories help identify which types of content trend more frequently, such as entertainment, gaming, music, education, or sports.

Before analysis, the dataset is cleaned and preprocessed. Missing values in columns like description and tags are handled properly to improve data quality. New features such as publish hour, publish day, and engagement rate are also created to make the analysis more meaningful. The dataset is used for both data analysis and machine learning tasks. Visualization techniques like bar charts, histograms, scatter plots, and heatmaps are used to identify patterns and relationships between different features. Machine learning models are then applied to predict video views and trending performance.

DATA EXPLORATION

The dataset used in this project is a large collection of YouTube trending video records. It contains 268,787 entries and 20 columns, which makes it suitable for deep analysis and machine learning tasks. The dataset includes both numerical and categorical features, allowing different types of analysis to be performed. The main columns include video_id, title, channelTitle, categoryId, trending_date, publish_time, tags, view_count, likes, dislikes, comment_count, and description. These fields help in understanding both the content of the videos and their performance on YouTube. In addition, there are derived or processed features such as publish_hour, publish_day, engagement_rate, and trending status, which help in analyzing user interaction patterns. From the summary statistics, it can be observed that engagement metrics like views, likes, and comments vary widely, showing that some videos perform extremely well compared to others. The dataset also shows that most columns have very few missing values, meaning the data is clean and reliable for analysis. The presence of boolean fields like comments_disabled and ratings_disabled helps in understanding user interaction restrictions on videos.

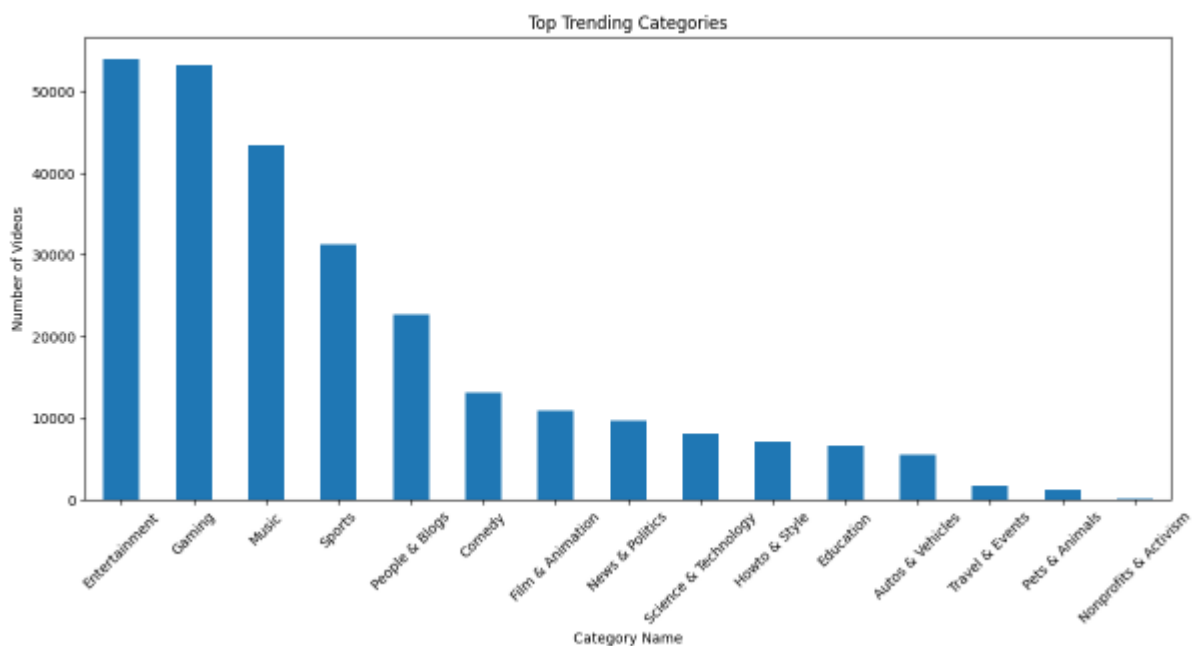
DATA CLEANING AND FEATURE ENGINEERING

In this step, the dataset was cleaned and prepared for analysis. First, missing values were checked to make sure the data was complete. Missing descriptions were replaced with “No Description” and missing tags were replaced with “No Tags”.

After cleaning the data, the publish time column was converted into a date and time format. New features were then created from it, including the publishing hour and publishing day. These features help in analyzing how upload timing affects video popularity.

An additional feature called **engagement rate** was also created using likes and comments compared to view count. This helps measure how actively viewers interact with videos and shows the relationship between engagement and popularity.

CATEGORY ANALYSIS

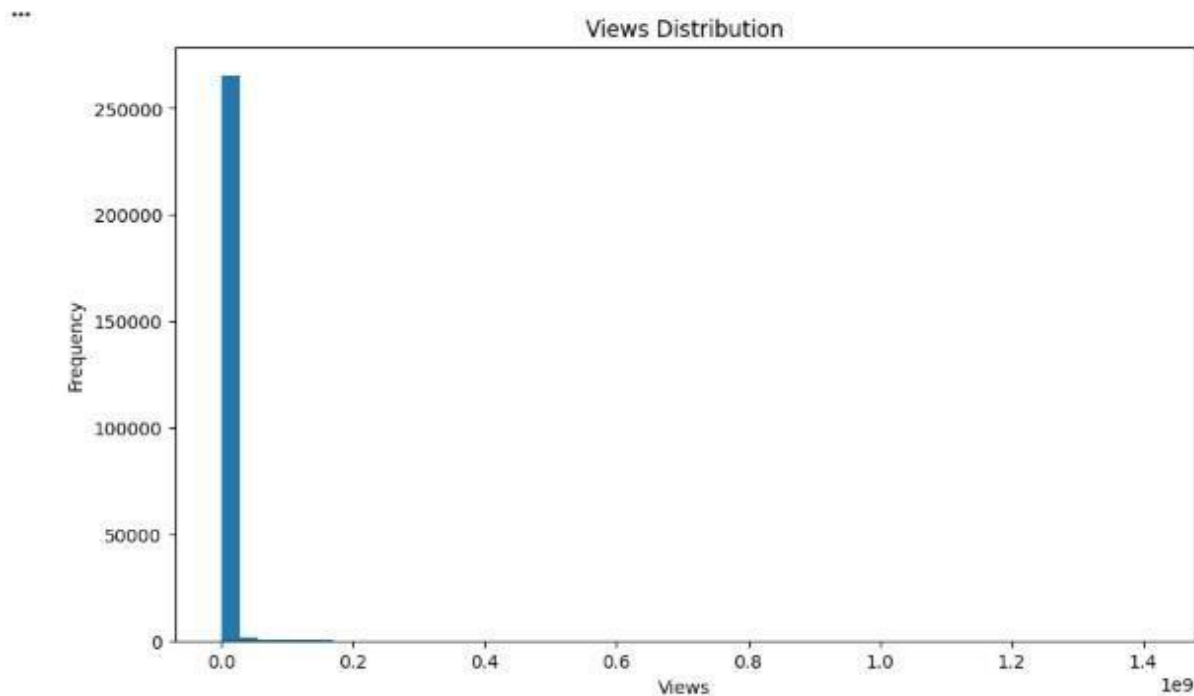


EXPLANATION

The bar chart displays the number of trending YouTube videos across different categories, with the x-axis representing the categories and the y-axis showing the total number of trending videos in each. From the graph, it is clear that **Entertainment** has the highest number of trending videos, followed closely by **Gaming**, indicating that users are most engaged with entertainment-based and gaming content on YouTube. The **Music** category also shows a strong presence with a large number of trending videos, highlighting its consistent popularity among viewers. Categories such as **Sports** and **People & Blogs** fall in the middle range, showing a moderate level of audience interest and engagement. In contrast, categories like **Comedy**, **Film & Animation**, **News & Politics**, and **Science &**

Technology have comparatively fewer trending videos. The lowest numbers are observed in **Travel & Events**, **Pets & Animals**, and **Nonprofits & Activism**, which suggests either lower upload frequency or reduced viewer engagement in these areas. Overall, the chart clearly indicates that entertainment-focused content dominates YouTube trends, while educational, niche, or less mainstream categories tend to appear less frequently in trending lists.

VIEWS DISTRIBUTION

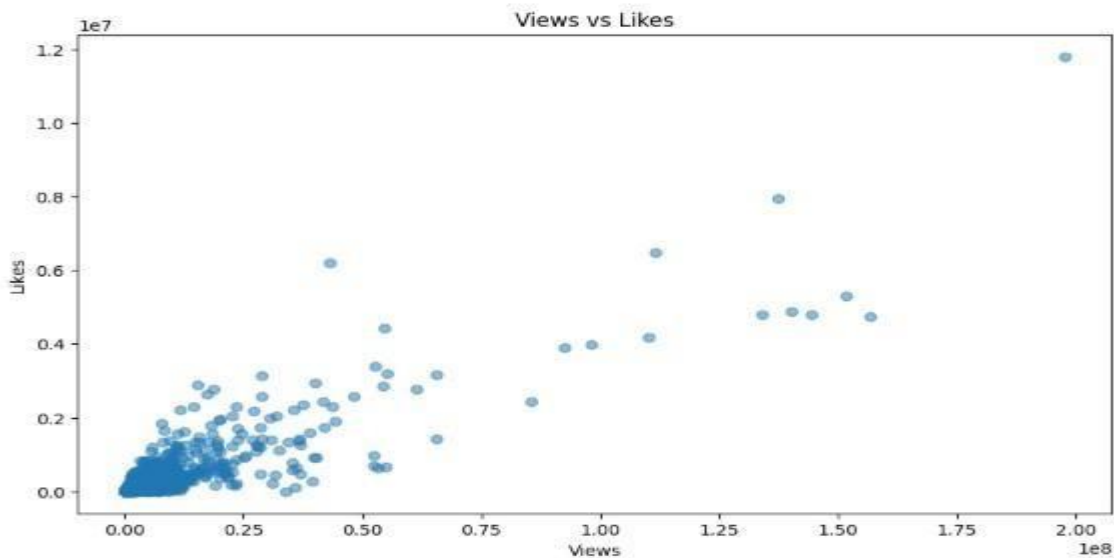


EXPLANATION

The views distribution histogram illustrates how video views are spread across the dataset. The x-axis represents the number of views while the y-axis represents the frequency of videos. The graph shows that most videos are concentrated near lower view counts, while only a small number of videos achieve extremely high views. This creates a right-skewed distribution where the majority of videos receive average or low popularity and only a few become viral. The long tail on the graph represents highly successful videos that gain massive audience attention through factors such as strong thumbnails, attractive titles, trending topics, celebrity influence, or algorithm recommendations. This analysis highlights the highly competitive nature of YouTube, where viral success is rare compared to normal performance levels. The

graph is also important from a machine learning perspective because extreme outlier videos can affect prediction accuracy and may require preprocessing techniques such as normalization or log transformation. Overall, the histogram demonstrates that YouTube popularity is unevenly distributed, with a small percentage of videos generating the majority of views and engagement on the platform.

VIEWS VS LIKES RELATIONSHIP

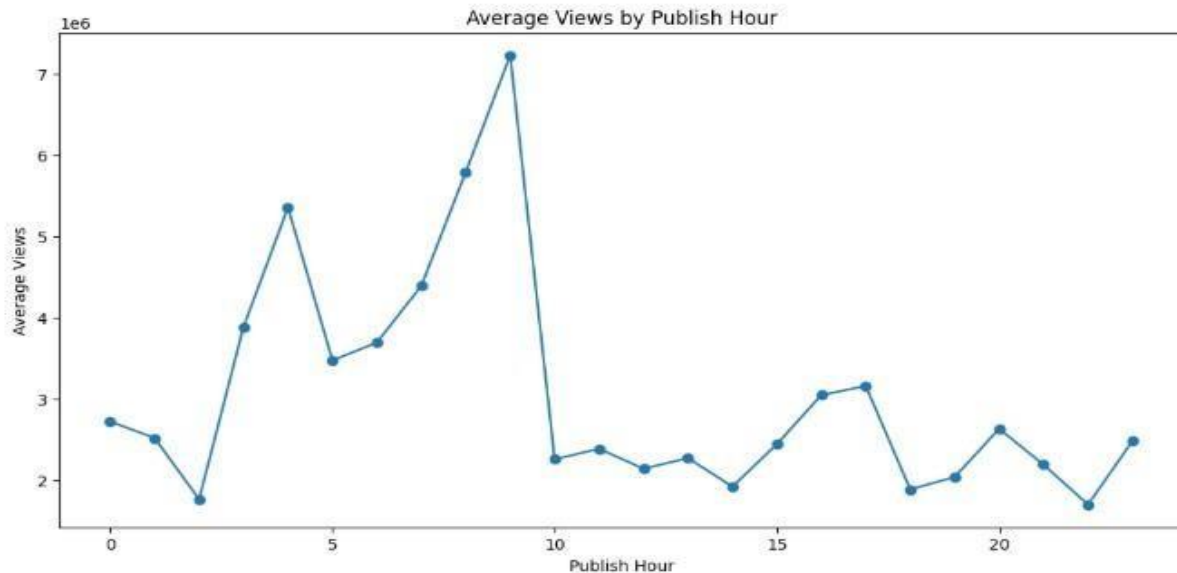


EXPLANATION

The Views vs Likes scatter plot illustrates the relationship between video views and the number of likes received. In the graph, the x-axis represents views while the y-axis represents likes. Each point corresponds to an individual video in the dataset. The graph clearly shows a positive relationship between the two variables, meaning videos with higher views generally receive more likes. This relationship occurs because a larger audience creates more opportunities for viewers to interact with the content. Most points are concentrated near lower view and like values, showing that the majority of videos receive moderate engagement, while only a few videos achieve extremely high popularity. The spread of the points also indicates that not all highly viewed videos receive the same number of likes, suggesting that audience satisfaction and content quality vary across videos. Some videos may gain views through curiosity or trends but may not necessarily generate strong positive reactions. This scatter plot is important because it visually confirms the correlation between popularity and

audience engagement. It also supports the machine learning model by showing that likes can serve as a strong predictor for estimating video views. Overall, the graph demonstrates that audience interaction and popularity are closely connected in YouTube video performance.

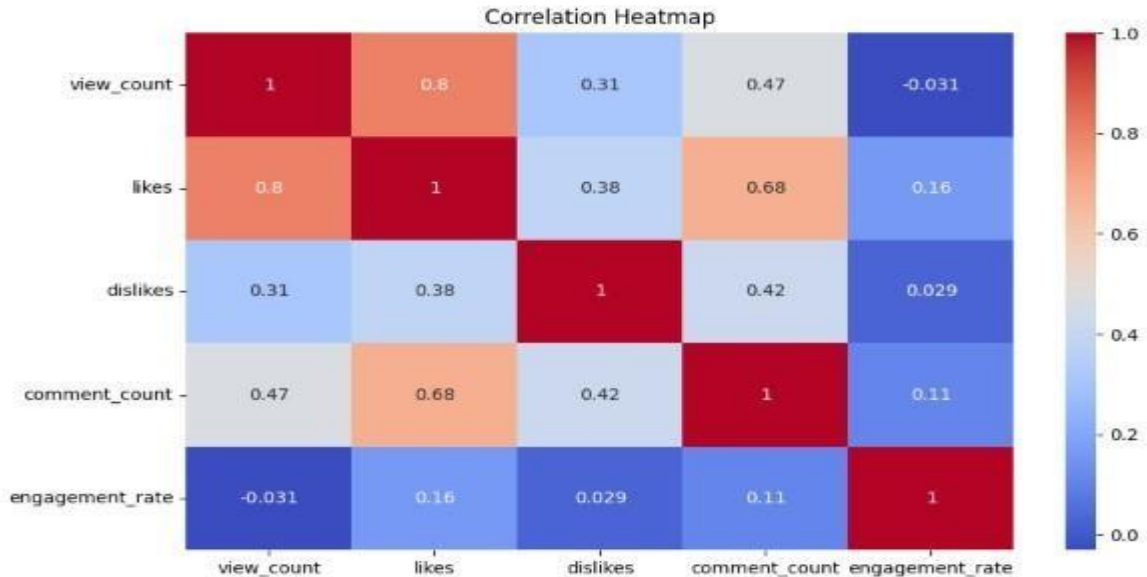
PUBLISH TIME ANALYSIS



EXPLANATION

The publish time analysis graph examines how the upload timing of videos affects average views. The x-axis represents the publish hour, while the y-axis represents average views received during those hours. The graph clearly shows that certain times of the day receive significantly higher average views compared to others. This indicates that audience activity changes throughout the day and upload timing plays a major role in video performance. During peak hours, more users are active on YouTube, allowing videos to gain faster engagement such as views, likes, and comments. Early engagement is important because YouTube's recommendation algorithm promotes videos that quickly attract audience interaction. Some hours in the graph show low average views, likely because viewers are inactive or sleeping during those times. The analysis demonstrates that creators can improve their chances of reaching larger audiences by uploading videos during high-traffic periods. This information is useful for content scheduling strategies and helps explain how timing contributes to trending success on YouTube.

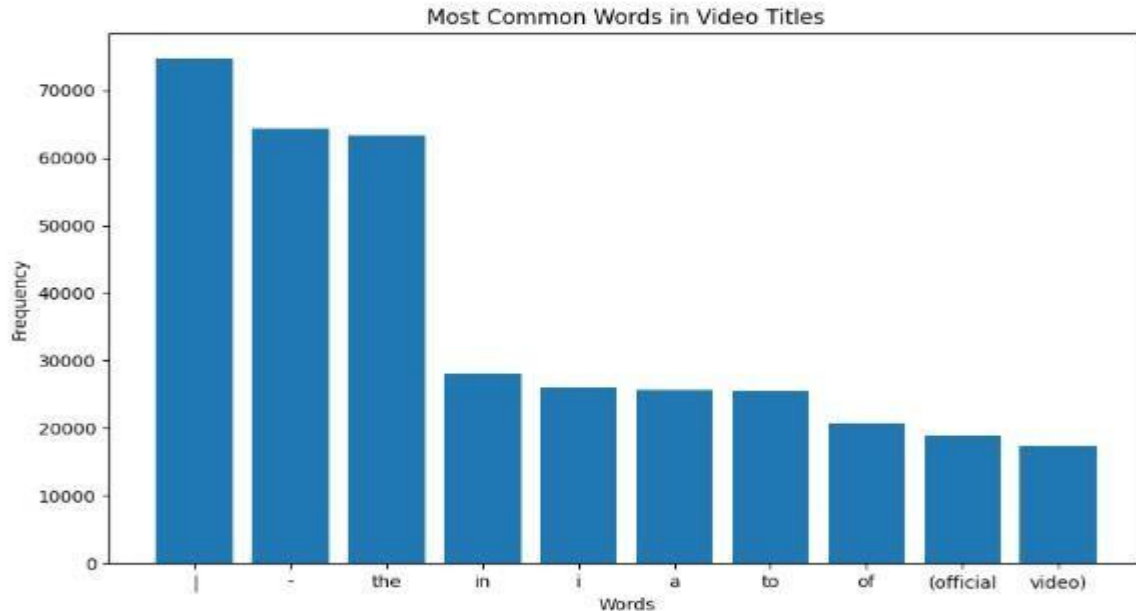
CORRELATION HEATMAP



EXPLANATION

The correlation heatmap shows the relationships between important numerical variables such as views, likes, dislikes, comments, and engagement rate. Correlation values range from -1 to +1, where values closer to +1 indicate a strong positive relationship. The heatmap reveals a very strong positive correlation between views and likes, meaning videos with higher views usually receive more likes because a larger audience interacts with them. There is also a strong relationship between likes and comments, which suggests that engaging videos encourage viewers to participate and share opinions. The moderate relationship between views and comments indicates that although popular videos often receive more comments, audience discussion also depends on the content type. The heatmap further shows that highly viewed videos may receive both likes and dislikes due to wider audience exposure. Engagement rate has weaker correlations with some variables, meaning high views alone do not always guarantee strong audience interaction. Overall, the heatmap is important because it identifies which features influence video popularity and helps in selecting useful variables for machine learning prediction models.

TITLE ANALYSIS



EXPLANATION

The title analysis graph represents the most commonly used words in YouTube video titles. The x-axis shows the words while the y-axis shows their frequency in the dataset. Words such as “I”, “the”, “in”, “to”, “of”, “official”, and “video” appear most frequently, which indicates that trending videos often use simple, direct, and engaging language. The high frequency of the word “I” suggests that many trending videos are personal, vlog-based, reaction-oriented, or storytelling content where creators directly connect with the audience. Similarly, words like “official” and “video” are commonly associated with music videos, movie trailers, announcements, and professionally branded uploads that attract large audiences. The analysis also shows that successful YouTube titles are usually short, easy to understand, and optimized to increase click-through rates. Since titles are the first thing viewers notice, creators carefully choose words that create curiosity and emotional engagement. This analysis helps explain how title wording contributes to video popularity and trending performance on YouTube.

MACHINE LEARNING MODELS AND RESULT

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LINEAR REGRESSION RESULTS  
MAE: 1374773.6356446634  
R2 Score: 0.5476463395664074  
  
RANDOM FOREST RESULTS  
MAE: 839785.0562590834  
R2 Score: 0.6522574328886642  
  
LOGISTIC REGRESSION RESULTS  
Accuracy: 0.8165296327988393
```

In this part of the project, machine learning models were used to predict YouTube video performance using features such as likes, dislikes, and comment count. The main goal was to analyze whether audience engagement data can help estimate video views and trending status. Different machine learning algorithms were applied to compare their prediction performance and understand which model works better for YouTube analytics.

The first step was preparing the machine learning data. Important features including likes, dislikes, and comment count were selected as input variables because these metrics directly represent audience interaction. The target variable was view count, which the models tried to predict. After selecting features and target values, the dataset was divided into training data and testing data using the train-test split method. The training data was used to teach the model patterns from the dataset, while the testing data was used to evaluate prediction accuracy on unseen data.

A Logistic Regression model was also created to predict trending status. In this approach, a new binary column called “trending” was generated where videos with views greater than the median view count were marked as trending videos. Logistic Regression was then trained to classify whether a video is trending or non-trending based on engagement metrics. The accuracy score produced by the model measured how correctly the algorithm classified videos into trending categories. The project also used a Linear Regression model to predict video views. Linear Regression works by identifying linear relationships between independent variables and the target variable. The model was trained using the selected features and then used to predict views for testing data. Two important evaluation metrics were used for performance measurement: Mean Absolute Error (MAE) and R^2 Score. MAE

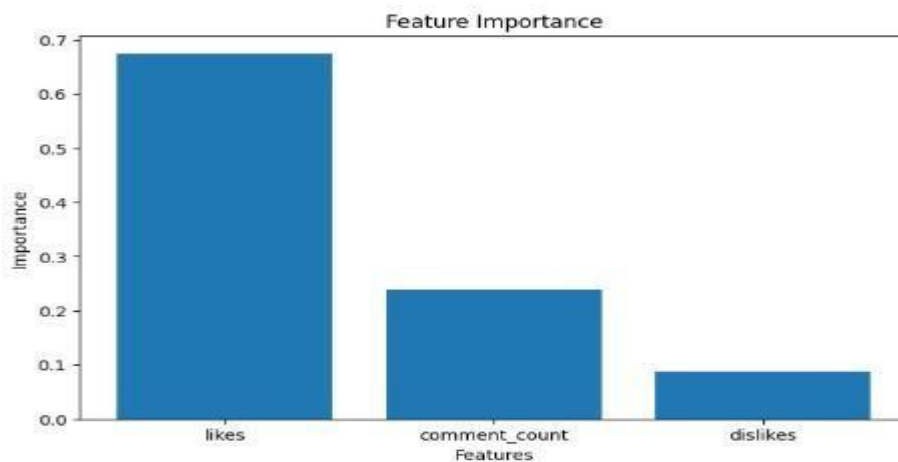
measures the average prediction error, while the R^2 Score measures how well the model explains variation in the data. A higher R^2 value indicates better prediction performance.

Another important model used in the project was Random Forest Regression. This model is more advanced than Linear Regression because it combines multiple decision trees to improve prediction accuracy and reduce overfitting. The Random Forest model was trained on the same features and then used to generate predictions for view counts. The results showed MAE and R^2 scores for model evaluation. Compared to Linear Regression, Random Forest generally performs better because it can capture complex relationships between variables instead of only linear patterns.

The results section displays the performance values of both models. The MAE values indicate the average difference between actual views and predicted views, while the R^2 scores indicate how accurately the model explains the data patterns. The project findings suggest that Random Forest Regression provides stronger prediction capability because YouTube data contains non-linear and complex relationships between audience engagement metrics and popularity.

Overall, this machine learning section demonstrates how data science techniques can be applied to social media analytics. By using Linear Regression, Random Forest Regression, and Logistic Regression, the project successfully analyzed YouTube engagement patterns, predicted video views, and classified trending videos. This part is important because it transforms raw data into predictive insights and helps understand the factors that influence YouTube video success.

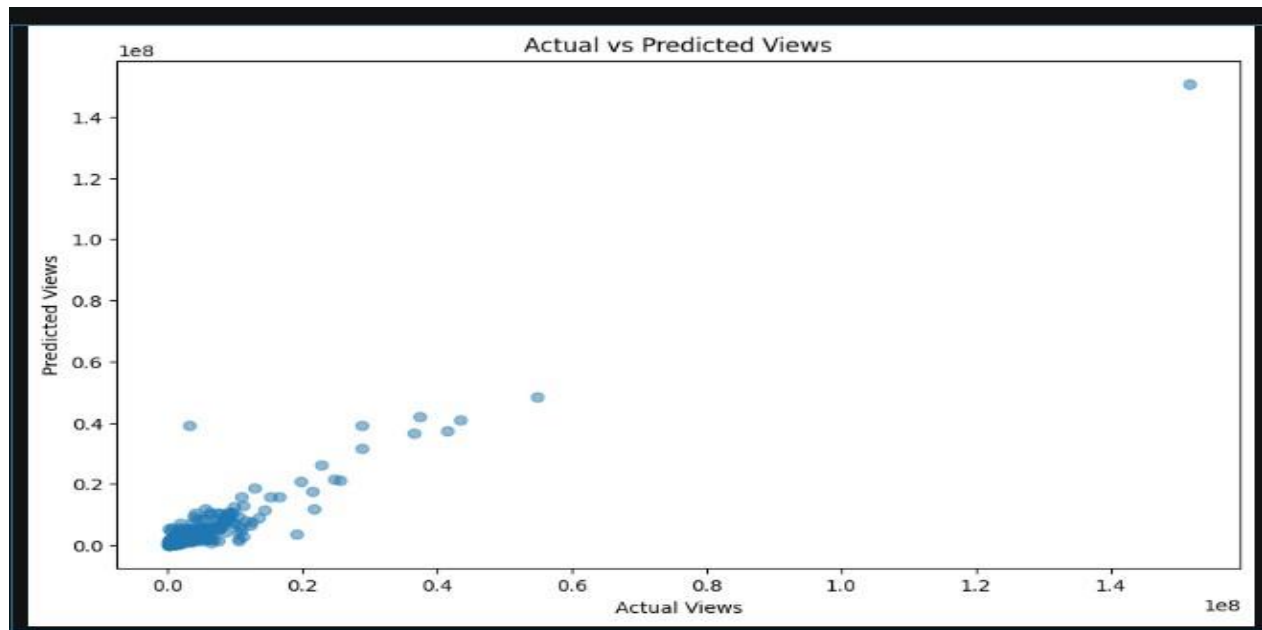
FEATURE IMPORTANCE



EXPLANATION

The feature importance graph shows which variables contributed most to the Random Forest model's predictions. The x-axis represents different features such as likes, comment count, and dislikes, while the y-axis represents the importance score assigned by the model. According to the graph, likes have the highest importance value, meaning they are the strongest predictor of video views and popularity. Comment count also contributes significantly, while dislikes have a much smaller effect compared to likes and comments. This analysis indicates that positive audience interaction plays a major role in determining whether a video becomes popular. The graph helps explain how the machine learning model makes decisions by identifying which features carry the most predictive power. In YouTube analytics, feature importance is very useful because it allows researchers and content creators to understand which engagement metrics influence performance the most. For example, if likes strongly affect views, creators may focus on improving audience satisfaction and encouraging interaction. The feature importance graph also supports the correlation analysis results, where likes and comments showed strong relationships with views. Overall, this graph demonstrates that audience engagement metrics are critical factors in predicting YouTube video success.

ACTUAL VS PREDICTED VIEWS



EXPLANATION

The Actual vs Predicted Views scatter plot is used to evaluate the performance and accuracy of the machine learning model. In this graph, the x-axis represents the actual video views from the dataset, while the y-axis represents the predicted views generated by the Random Forest model. Each point on the graph represents one video. If the model predictions were perfectly accurate, all points would lie exactly on a straight diagonal line where predicted values equal actual values. The graph shows that many points cluster close to the expected trend, indicating that the model is able to identify general patterns in the data. However, some points are spread farther away from the ideal alignment, especially for videos with extremely high views. This suggests that the model performs better on average-view videos but has more difficulty predicting viral or highly popular videos. Such prediction variation is common because viral content is influenced by unpredictable factors such as trends, audience emotions, social media sharing, and recommendation systems. Despite some prediction errors, the scatter plot demonstrates that the model successfully captures the overall relationship between input features and video popularity. This visualization is important because it helps measure machine learning effectiveness and reveals strengths and weaknesses in prediction accuracy.

PREDICTION TABLE

FIRST 5 RANDOM FOREST PREDICTIONS		
	Actual Views	Predicted Views
0	546263	1106611.64
1	589224	557629.58
2	249016	536767.90
3	133954	335165.05
4	974385	1056294.74

EXPLANATION

This table shows the first five predictions generated by the Random Forest Regression model. The purpose of this model is to predict YouTube video views based on input features such as likes, comments, dislikes, engagement metrics, and other related variables. The table compares the “Actual Views” from the dataset with the “Predicted Views” generated by the machine learning algorithm. By comparing these two values, we can evaluate how accurately the model is able to estimate video popularity. In some cases, the predicted views are very close to the actual views, which indicates good model performance. However, in other cases there is a noticeable difference between actual and predicted values, showing that prediction errors still exist. These differences occur because YouTube popularity depends on many unpredictable factors such as trends, audience behavior, recommendation algorithms, thumbnails, and content quality that cannot be fully captured by numerical data alone. The Random Forest model works by combining multiple decision trees and averaging their outputs, which improves prediction accuracy and reduces overfitting compared to a single decision tree. This table is important because it provides a practical demonstration of how machine learning can be used to estimate future video performance using historical YouTube data.

COMPARISON TABLE

Comparison Point	Linear Regression	Random Forest	Logistic Regression
Purpose	Predict video views	Predict video views	Predict trending status
Metrics Used	MAE, R ² Score	MAE, R ² Score	Accuracy
Performance	Moderate	Best	Good
Ability to Handle Complex Data	Low	High	Moderate

Main Advantage	Simple and fast	Most accurate and reliable	Effective for classification
Main Limitation	Lower accuracy on complex data	More computational time	Cannot predict exact views
Best Use in Project	Baseline comparison	View prediction	Trending prediction
Overall Result	Acceptable	Best overall model	Best classification model

Which Model Performed Best?

Best Overall Model: Random Forest

Why?

- It gave:
 - **Lower MAE**
 - **Higher R² Score**
- It handles:
 - complex patterns
 - feature interactions
 - non-linear relationships

In this project:

YouTube popularity is influenced by many combined factors:

- likes
- comments
- engagement
- publishing behavior

These relationships are not perfectly linear, so Random Forest performs better.

Why Linear Regression Was Weaker

Linear Regression assumes a straight-line relationship between variables.

But YouTube data is more complex:

- some videos suddenly go viral
- engagement grows non-linearly
- categories behave differently

So Linear Regression could not model those patterns as effectively.

Why Logistic Regression Was Used

Logistic Regression was not used for predicting exact views.

Instead, it was used for:

- predicting whether a video is “Trending” or “Not Trending”

So it solves a different problem:

- classification instead of regression

That’s why its metric is:

- Accuracy instead of MAE/R²

CONCLUSION

This project successfully analyzed YouTube trending videos using data analysis, visualization, and machine learning techniques to understand the factors that influence video popularity and trending behavior. By using a large dataset of YouTube videos, the project identified important relationships between views, likes, comments, categories, and publish time.

The analysis showed that audience engagement plays a major role in YouTube success. Videos with higher likes and comments generally received higher views, proving that audience interaction strongly affects popularity. The correlation heatmap and feature importance analysis confirmed that likes were the strongest predictor of video performance, followed by comments. The project also showed that YouTube popularity is unevenly distributed, where only a small number of videos become highly viral while most videos receive average views.

Another important finding was the effect of upload timing on video performance. Videos published during peak audience activity hours received better engagement and higher average views. Title analysis also revealed that trending videos often use simple, engaging, and attention-grabbing words to attract viewers and improve click-through rates.

Machine learning models were successfully applied to predict video views and trending status. Linear Regression provided moderate prediction results but

struggled with complex patterns in YouTube data. Random Forest Regression performed best because it handled non-linear relationships and feature interactions more effectively. Logistic Regression also achieved good accuracy in classifying videos as trending or non-trending.

Overall, the project demonstrated how data science and machine learning can be used in social media analytics to understand audience behavior and predict content performance. The findings can help content creators and marketers improve upload strategies, increase audience engagement, and optimize video performance on YouTube. Future improvements could include sentiment analysis, thumbnail analysis, and deep learning models for even more accurate predictions.